MCS-214	MACHINE LEARNING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To introduce students to the basic concepts and techniques of Machine Learning.
CO2	To develop skills of using recent machine learning software for solving practical problems.
CO3	To gain experience of doing independent study and research.
CO4	To Learn about Support Vector Machine concepts.
CO5	To learn about clustering

MODULE I

Algorithmic models of learning. Learning classifiers, functions, relations, grammars, probabilistic models, value functions, behaviors and programs from experience. Bayesian, maximum a posteriori, and minimum description length frameworks. Parameter estimation, sufficient statistics, decision trees, neural networks, support vector machines, Bayesian networks, bag of words classifiers,

MODULE II

N-gram models; Markov and Hidden Markov models, probabilistic relational models, association rules, nearest neighbor classifiers, locally weighted regression, ensemble classifiers. Computational learning theory, mistake bound analysis, sample complexity analysis, VC dimension, Occam learning, accuracy and confidence boosting. Dimensionality reduction, feature selection and visualization.

MODULE III

Clustering, mixture models, k-means clustering, hierarchical clustering, distributional clustering. Reinforcement learning; Learning from heterogeneous, distributed, data and knowledge. Selected applications in data mining, automated knowledge acquisition, pattern recognition, program synthesis, text and language processing, internet-based information systems, human-computer interaction, semantic web, and bioinformatics and computational biology.

References

1. *Bishop, C. (2006). "Pattern Recognition and Machine Learning". Berlin: Springer-Verlag.*

Course Outcome:

CO1	Explain Machine Learning concepts, classifications of Machine Learning and write simple programs using python.
CO2	Describe Supervised Learning concepts
CO3	Explain Support Vector Machine concepts
CO4	Describe unsupervised learning concepts and dimensionality reduction techniques
CO5	Discuss simple Machine Learning applications in a range of real-world applications using Python programming